

Category Theory and Functional Programming

Universality and Adjoints

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Preliminaries

Convention

The number assigned to chapters, examples, exercises, figures, pages, sections, and theorems on these slides correspond to the numbers assigned in the textbook (Abramsky and Tzevelekos 2011).

Outline

Adjunctions for Pre-orders

Universal Arrows

Exponentials

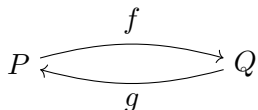
Cartesian Closed Categories

References

Adjunctions for Pre-orders

Definition

Let $(P, \preceq_P)$ and $(Q, \preceq_Q)$ be two pre-ordered sets and let f and g be two homomorphisms (monotone maps) forth and back.



The maps f and g are **adjoint**, denoted $f \dashv g$, when for all $x \in P$, $y \in Q$,

$$f x \preceq_Q y \quad \text{iff} \quad x \preceq_P g y.$$

The map f is the **left adjoint** and the map g is the **right adjoint** (Awodey and Bauer 2022).

Universal Arrows

Universal Arrows

Definition

Let $G : \mathcal{D} \rightarrow \mathcal{C}$ be a functor and let C be an object of $\mathcal{C}$. A **universal arrow** from C to G is a pair (D, η) where D is an object of $\mathcal{D}$ and η is an arrow

$$\eta : C \rightarrow G_0 D$$

Exponentials

Exponentials

Introduction

We shall introduce abstract characterisations of the following facts of set theory:

- (i) The functions of a **set** X to a **set** Y , denoted Y^X , are also a **set**.
- (ii) A function of **two** variables $f : X \times Y \rightarrow Z$ can be represented as a function of **one** variable $\lambda f : X \rightarrow Z^Y$.

Exponentials

Set theory

Let Z^Y be the set of functions from the set Y to the set Z . For any set X and function $g : X \times Y \rightarrow Z$, the following diagrams commutes:

$$\begin{array}{ccc} & Z^Y & \\ \uparrow \Lambda g & & \\ X & & \end{array} \quad \begin{array}{ccc} & Z^Y \times Y & \xrightarrow{\text{eval}_{Y,Z}} Z \\ \uparrow \Lambda g \times \text{id}_Y & \nearrow g & \\ X \times Y & & \end{array}$$

$$(g = \text{eval}_{Y,Z} \circ (\Lambda g \times \text{id}_Y))$$

where

$$\text{eval}_{Y,Z} : (Z^Y \times Y) \rightarrow Z := (f, y) \mapsto f y$$

(application),

$$\Lambda g : X \rightarrow (Y \rightarrow Z) := x y \mapsto g(x, y)$$

(currying).

Exponentials

Definition

Let $\mathcal{C}$ be a category with binary products, and let Z and Y be objects of $\mathcal{C}$. The pair (Z^Y, eval) , where Z^Y is an object of $\mathcal{C}$ and eval is an arrow

$$\text{eval} : (Z^Y \times Y) \rightarrow Z$$

is an **exponential object** iff for any object X and arrow $g : X \times Y \rightarrow Z$, there is a **unique** arrow $\Lambda g : X \rightarrow Z^Y$ such that the following diagram commutes:

The diagram illustrates the universal property of the exponential object Z^Y . It consists of the following objects and arrows:

- Object X at the bottom left.
- Object $X \times Y$ at the bottom middle.
- Object Z^Y at the top left.
- Object $Z^Y \times Y$ at the top middle.
- Object Z at the top right.

The arrows are:

- A vertical dotted arrow from X to Z^Y labeled Λg .
- A vertical dotted arrow from $X \times Y$ to $Z^Y \times Y$ labeled $\Lambda g \times \text{id}_Y$.
- A diagonal solid arrow from $X \times Y$ to Z labeled g .
- A horizontal solid arrow from $Z^Y \times Y$ to Z labeled $\text{eval}_{Y,Z}$.

The diagram commutes, meaning $g = \text{eval}_{Y,Z} \circ (\Lambda g \times \text{id}_Y)$.

$$(g = \text{eval}_{Y,Z} \circ (\Lambda g \times \text{id}_Y))$$

Cartesian Closed Categories

Cartesian Closed Categories

Definition

A category is a **Cartesian closed category** iff it has finite products (i.e. a terminal object and binary products) and exponentials.

References

References



S. Abramsky and N. Tzevelekos (2011). Introduction to Categories and Categorical Logic. In: New Structures for Physics. Ed. by Bob Coecke. Vol. 813. Lecture Notes in Physics. Springer, pp. 3–94. DOI: [10.1007/978-3-642-12821-9_1](https://doi.org/10.1007/978-3-642-12821-9_1) (cit. on p. 2).



Steve Awodey and Andrej Bauer (2022). Introduction to Categorical Logic. Appendix A. Category Theory. Draft version 2022-01-15. URL: <https://awodey.github.io/catlog/> (cit. on p. 4).